TASK 1: FRONT WHEEL ALIGNMENT AND STEERING GEOMETRY CHECKS

Task reference: Block_4_Task_1_Front_Wheel_Alignment

Task overview

Block
Block 4 - Emerging Technology and Driveline Systems

Weeks
Weeks 31-40

Duties covered
1, 3, 4, 5, 6, 7, 8, 9, 10

KSBs covered
K16-K20, S10-S13, B1, B2

Estimated duration
60 minutes

Assessment type
Practical / knowledge / documentation

Scenario

A customer reports uneven tyre wear and the steering wheel is off-centre. You must carry out alignment pre-checks, record geometry readings and explain whether adjustment or further repair is required.

Objectives

- Carry out wheel alignment pre-checks safely.
- Record camber, caster and toe readings where equipment allows.
- Identify whether readings are within manufacturer specification.
- Explain the relationship between geometry and tyre wear.
- Record findings and customer recommendations.

Materials you need at the workstation

- Light vehicle
- Wheel alignment equipment
- Tyre pressure gauge and tread depth gauge
- Manufacturer geometry data
- PPE and Apprentice Assessment Record

Complete all relevant fields. Enter N/A where a field does not apply.

Practical task instructions

1. Prepare the vehicle and alignment equipment safely.
2. Carry out pre-checks including tyre pressure, tyre condition and suspension condition.
3. Measure alignment readings and compare with manufacturer data.
4. Adjust if instructed and within competence, or explain referral/action required.
5. Record findings and reinstate the vehicle/workstation.

Vehicle or equipment details

Vehicle make

Mileage / hours

Vehicle model

Engine size / system type

Registration / fleet number

VIN / chassis / equipment ID

Work completed and safe working

Brief description of work carried out and any further recommendations

PPE worn and special precautions taken

Technical data and measurements

Front toe measured

Front toe specification

Camber measured

Camber specification

Caster measured

Rear toe/thrust angle

Tyre pressures

Ride height/pre-check result

Special tools including any required calibration

Knowledge questions

1. Why are tyre pressures checked before alignment?

2. What effect can incorrect toe have on tyre wear?

3. Why should suspension condition be checked before adjustment?

Knowledge questions continued

4. What does thrust angle indicate?

5. How would you explain alignment findings to a customer?

6. When should alignment not be adjusted immediately?

7. What evidence should be retained from the alignment task?

Assessor observation notes

Record observations, evidence reviewed and any areas requiring further development

TASK 1: BLOCK 4 APPRENTICE ASSESSMENT

Declarations, sign-off and assessment decision

Assessment declarations

Apprentice declaration: I confirm that the work recorded in this assessment document is my own work and that I completed the task under the conditions stated.

Assessor declaration: I confirm that I have observed and/or reviewed the evidence recorded in this document and that the assessment decision is accurate.

Apprentice sign-off

Apprentice name

Date signed

Apprentice signature

Assessor sign-off

Assessor name

Assessor PIN

Date signed

Assessor signature

Sign using your PDF reader's signing tool, or sign by hand after printing.

Assessment decision and validation

Assessment decision - select Pass or Refer; tick resubmission if required

- [ ] Pass
- [ ] Refer
- [ ] Resubmission required

Assessment / evidence reference

Assessor feedback and action points